3rdParty MCAL Integration

Technical Reference

NXP S32K14x

Version 1.5.0

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Status	Released

Document Information

History

Author	Date	Version	Remarks
Andrej Gazvoda	2017-07-19	1.0.0	Basic Integration of S32K14x MCAL
Roland Suess	2017-08-11	1.1.0	Added MixedASR Integration (ASR4 MCAL in MSR3 SW Stack)
Andrej Gazvoda	2017-09-20	1.2.0	Basic Integration of AUTOSAR 4.2.2 MCAL for S32K14x
Andrej Gazvoda	2017-10-26	1.2.1	Added AUTOSAR 4.2.2 CSEC package
Roland Suess	2017-12-20	1.2.2	Adapted chapter 1.4.1 (Distinguish Vector Tools)
Marco Ziegler	2018-05-29	1.3.0	Renamed to Technical Reference naming scheme Added three known issues in MCAL Package and the hotfix they were fixed
Kishore Gunda	2018-08-14	1.4.0	MIP update for MCAL S32K14X_MCAL4_2_RTM_HF1_1_0_1
Kishore Gunda	2018-08-28	1.5.0	Include I2C module to MIP, update the MCAL with the latest HotFix S32K14X_MCAL4_2_RTM_HF3_1_0_1

Reference Documents

No.	Source	Title	Version
[1]	Vector	TechnicalReference_3rdParty-MCAL-Integration.pdf	See Delivery
[2]	Vector	ScreenCast_McalIntegration_Tresos.pdf	See Delivery

Scope of the Document

This document contains information about the integration of 3rd Party MCAL into Vector software stack.

Contents

1 MCAL Integration

1.1 Type of Integration

Basic Integration
Both configuration tools, EB tresos™ as well as DaVinci Configurator, are used for configuration.

Recommended workflow:
Start initial configuration with EB tresos™, export it in AUTOSAR format and import it into DaVinci Configurator. Generation and minor changes in configuration are done in DaVinci Configurator.
For usage with DaVinci Configurator please refer to chapter 'Mixed configuration tool usage' in the document `TechnicalReference_3rdParty-MCAL-Integration.pdf` [1].

Furthermore, you will find a Multimedia Link in the document `ScreenCast_McalIntegration_Tresos.pdf` [2].

1.2 MCAL Location within SIP

The 3rd Party MCAL is separated from the Vector parts within the SIP. Furthermore, it might not be part of the delivery. Please refer to chapter 'First Steps' in document `TechnicalReference_3rdParty-MCAL-Integration.pdf` [1].

1.3 Supported 3rd Party Products

This integration supports the following NXP targets:

- S32K14x

Note
Please refer to the Release Notes of the 3rd Party Products for further information, e.g. regarding supported versions, derivatives and compilers.

Note
Please be aware that only official 3rd Party vendor releases are part of this Vector integration package. Therefore any customer-specific releases cannot be considered.

Caution
Please contact the 3rdPartyVendor to find out if there are further Hotfixes available for your Mcal package.
It is essential to replace the affected module plugins in your original package before you start Script_MCAL_Prepare.bat.

1.4 Configuration Tools

1.4.1 S32K14x

- DaVinci Configurator 5.15 (and higher) for MSR4 projects
- DaVinci Configurator 4.2.3.2 (and higher) for MSR3 projects (MixedASR)

2 Vector Comment

Please consider the attached `TechnicalReference_3rdParty-MCAL-Integration.pdf` [1] for further information regarding Vector integration and setup of a project.

2.1 Known Issues

2.1.1 S32K14x MCAL Package ASR 4.0.3

2.1.1.1 Default configuration of Mcu PLL

The default configuration of Mcu (configuration parameter `McuNoPll`) deactivates the declaration of API function `Mcu_DistributePllClock`; this results in a linker error. For this reason, the MCAL integration provides proper parameter configuration with the Mcu recommended configuration. Please consider that this configuration will be ignored if another configuration gets imported to DaVinci Configurator, and must be updated by the user.

2.1.2 S32K14X_MCAL4_2_RTM_1_0_1

2.1.2.1 Default configuration of Mcu PLL

The default configuration of Mcu (configuration parameter `McuNoPll`) deactivates the declaration of API function `Mcu_DistributePllClock`; this results in a linker error. For this reason, the MCAL integration provides proper parameter configuration with the Mcu recommended configuration. Please consider that this configuration will be ignored if another configuration gets imported to DaVinci Configurator, and must be updated by the user

2.1.3 S32K14x Security Package ASR 4.2.2

2.1.3.1 S32K14X_CSEC_MCAL4_2_RTM_1_0_0 and S32K14X_CSEC_MCAL4_2_RTM_1_0_1: Error in Folder Structure

The CSEC package in S32K14X_CSEC_MCAL4_2_RTM_1_0_0 and S32K14X_CSEC_MCAL4_2_RTM_1_0_1, in folder ‘eclipse’ the folder ‘plugins’ is missing.

As a workaround, create the plugins folder and move the content of ‘eclipse’ into ‘plugins’ and run the helper tool.

This issue has been reported to NXP’s issue data base as Case 00175561

3 Glossary and Abbreviations

3.1 Glossary

Term	Description
3rd party components / MCAL	BSW modules not provided by Vector. Vector may have integrated the software within the SIP but does not take over any responsibility regarding functionality of these modules.
DaVinci Configurator	Configuration and generation tool for Vector MICROSAR components

Table 3-1 Glossary

3.2 Abbreviations

Abbreviation	Description
MCAL	Microcontroller Abstraction Layer
AUTOSAR	Automotive Open System Architecture
SIP	Software Integration Package (as provided by Vector)
Msn	Module Short Name according AUTOSAR
MSR	MICROSAR - Vector AUTOSAR Product

Table 3-2 Abbreviations

4 Contact

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